

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

Chapter 5

SUPPLY

DEFINITION OF SUPPLY:-

Supply of any good or service depends upon the behaviour of producers or sellers of that particular commodity or service.

Supply of a good cannot be discussed without its price.

Thus supply means the quantity of a commodity offered for sale in the market at a specific price during a given time period.

To understand the concept of supply of a good more comprehensively, following factors must be kept before:

(A) PRICE OF THE GOOD:

Quantity supplied is closely related with the price of the commodity. Any change in the price of the good influences its quantity supplied. For instance; at Rs.50 per K.G. the supply of sugar is 100 K.G. in the market. When its price rises to Rs.60 per K.G. quantity supplied moves upward to 200 K.G.. As an increase in price gives more profit to the suppliers and vice versa.

Hence $q_s = f(p)$

(B) AN INCREASING FUNCTION

As supply expands with the increase in price of a good and it contracts due to decrease in price. Hence quantity supplied is an increasing function of price.

(C) RESERVE PRICE:

It is the "minimum acceptable price" at which the seller is willing to sell the commodity. Below this price he can refuse to sell it. This price depends upon the nature of good and requirements of the supplier.

(D) PERIOD OF TIME:

According to it supply of a commodity may be assessed as under:-

(i) **Fixed Supply:** It is related with perishable goods e.g. Vegetables, Fruit, Fresh Milk, Fish etc. Their daily supply remains fixed in day to day market. It is known as market supply.

(ii) **Short period supply:** In short period supply of goods can be adjusted only up to a certain limit. It refers to the supply of goods available in the "fixed form" which is kept in stock, warehouse, Farm houses or at any place near the market. Excess capacity of the producing units may also help to increase supply. Thus in short period, supply curve becomes steeper and positively sloped.

(iii) **Long period supply:** In long period large supply adjustments are possible according to the changes in demand of the good. Time is sufficient to extend or contract the existing capacity and size of the producing units. New firms can also be installed and old one can be closed. This long period supply curve becomes "Flatter".

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DIFFERENCE BETWEEN SUPPLY AND STOCK

Stock represents the “total quantity” of a commodity available in shops, houses, farm houses, or warehouses near the market in “Finished form.” It can be brought for sale at a short notice.

Supply is the quantity of a commodity actually offered for sale in the market at a specific price level. Supply increases with the rise in price and vice versa. Normally supply is a part of the stock but it may become equal to the stock when whole quantity is sent in the market for sale.

Supply is always related with the certain price while stock has no price. Suppose a producer is having 1000 bags of wheat in the Godowns and farm houses. When he supplies 200 bags in the market for sale at the price of Rs.1200 per bag. Then stock = 1000 bags and supply = 200 bags (Here supply becomes a part of the stock)

Generally there is no remarkable difference between stock and supply of perishable goods e.g. Vegetables, Fruit, Fresh milk, Fish etc. As these goods have day-to-day supply in the market and whole stock must be sold otherwise it will be wasted. On the other hand if the commodity is durable e.g. wheat, sugar, rice etc. and the price offered is not acceptable, a part of the stock may be withheld. As price goes up, more quantity of durable good (wheat) out of the stock is offered for sale. Thus price plays an important role in this respect.

LAW OF SUPPLY

It is observed in the market that there is “direct and positive relationship” between price and supply of a commodity. It means that supply and price move in the same direction i.e. more supply at higher price and vice versa.

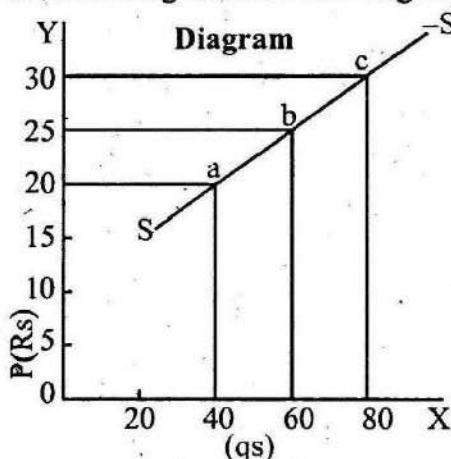
Thus $q_s = f(p)$ “quantity supplied is an increasing function of the price.”

Law of supply can be defined as under:-

“Other things remaining the same, quantity supplied of a commodity expands with an increase in price and contracts with the decrease in price”

Law of supply can be explained with the following schedule and diagram.

Schedule	
p(Rs)	qs(K.G)
Rs. 20	40
Rs. 25	60
Rs. 30	80



The above schedule shows three combinations of prices and quantity supplied in the market. Keeping in view these combinations we have achieved points a, b and c in the diagram. While connecting these points with each others, we arrive at SS supply curve. This

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curve is positively sloped moving upward to right showing a direct relationship between price and quantity supplied of the good.

SALIENT FEATURES OF THE SUPPLY CURVE

1. It indicates quantity supplied of a good offered for sale in the market at different prices during a given period of time.
2. Normally it is positively sloped moving upward from left to right.
3. Supply curve of perishable goods being inelastic becomes vertical upon x-axis.
4. Long period supply curve is more elastic (Flatter) than short period supply curve (steeper).

WHY SUPPLY CURVE IS POSITIVELY SLOPED?

1. PROFIT MOTIVE:

Profit increases with the rise in prices thus more quantity is supplied in market for sale.

2. DANGER OF FALL IN PRICE:

At a comparatively higher price more quantity is supplied to sell the stock immediately. There may be an anticipation of a fall in price and danger of loss in future. To avoid such a danger less quantity is withheld in the stock and more is supplied in the market.

3. INCREASE IN THE COST OF PRODUCTION:

When more units of the factors of production are employed to increase production. They may demand a rise in remunerations. It pushes cost of production upward. Which can only be faced when more units of commodity are supplied at higher prices.

ASSUMPTIONS OF THE LAW OF SUPPLY

This law may be proved true, in the presence of following assumptions.

1. NO CHANGE IN THE COST OF PRODUCTION:

It is assumed that cost of production remains the same, otherwise the law of supply may not prove true. If the cost of production rises, the supply may fall at the same price, because of a shortfall in profit. Similarly, at a lower cost of production related with a fall in price of a commodity may not be resulting of contraction in quantity supplied.

2. NO CHANGE IN PRODUCTION TECHNIQUES:

Production method should not change. Because the use of latest modern technology may decrease the cost of production. This may suspend the application of the law of supply. Because at the same price quantity supplied may rise.

3. NO CHANGE IN THE PRICE OF MACHINES AND RAW MATERIAL:

Next assumption of the law is that the prices of necessary tools, machines and raw material required for the production of good should remain constant. If these factors become cheaper, supply may not decrease even at the lower price and vice versa.

4. NO DISCOVERY OF NEW NATURAL RESOURCES:

Discovery of new mineral resources e.g. petrol, iron, coal, metals etc may result a fall in the cost of production of industrial and Agricultural products. Thus the supply of these products may increase even if price falls. Hence to prove the truth of the law of supply, it is assumed that no new natural resources are discovered.

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5. NO GOVERNMENT RESTRICTIONS:

No restrictions should be imposed upon the shifting and transportation of the good from one area to the other. As such restrictions do not allow the supply to increase in a market where price of the commodity has gone up.

6. POLITICAL STABILITY SHOULD PREVAIL:

When there are political disturbances and un-rest in the country quantity produced of the commodity will be decreased. Thus supply may fall even though price has not fallen.

LIMITATIONS AND EXCEPTIONS

Following are the limitations of the law of supply:

1. INTENSIVE NEED OF CASH MONEY:

When the producer or seller is badly in need of cash money. He may increase the supply of the commodity even though the price is at lower level. Thus, law of supply will lose its validity.

2. UN-CERTAIN CONDITIONS:

Danger of war, uncertainty, un-rest, floods, etc. may force the sellers to sell the stock of goods immediately even if the prices are low. Moreover, strikes by the workers or lock up by the owners, affects the production badly and supply cannot be increased even though prices are higher.

3. IMMEDIATE MIGRATION:

Due to unforeseen factors some time people are forced to migrate from one place to the other, or from one country to the other countries. Thus they increase the supply of their belongings even at the lower price. For instance, at the time of partition in 1947, immigrants sold their goods at a very low prices. The law of supply became ineffective under these conditions.

4. OBSTACLES IN THE WAY OF MEANS OF TRANSPORTATION:

When due to bad weather conditions, heavy snowfall, land sliding, floods, earthquakes etc. roads and railway tracks are blocked. It becomes very difficult to supply goods in the markets where prices are high. The law of supply is negated under these circumstances.

FUNCTIONAL EQUATION OF SUPPLY

According to the law of supply quantity supplied is an increasing function of price.

Thus we can say that Functional equation of supply exhibits a functional relationships between market price of a good and its quantity supplied for sale.

Profit margin also plays an important role in this respect.

$$\text{As } \pi = R - C$$

Here π = symbol of profit.

R = Total Revenue of the Supplier

C = Total cost of the commodity being supplied.

The sale plan of the firm or seller may be shown as under:

$$R = P_1 \cdot q_1 + P_2 \cdot q_2 + P_3 \cdot q_3 + \dots + P_n \cdot q_n$$

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- Here (i) R = Total revenue of the firm or seller
 (ii) $P_1 \dots P_n$ = The market prices of goods supplied.
 (iii) $q_1 \dots q_n$ = Quantities supplies of different goods.

On the other hand total cost of production depends upon the units of factors hired $\times$ Reward (price) given to these factors.

Thus purchase plan of the firm is:

$$C = px_1 \cdot x_1 + px_2 \cdot x_2 + px_3 \cdot x_3 + \dots + px_n \cdot x_n$$

C = Total Cost of Production
 $px_1 \dots px_n$ = Reward (prices) given to various factors hired
 $x_1 \dots x_n$ = units of various factors employed.

As the major determinants of the supply are:

- (i) P = Market price of the good
 (ii) C = Cost of production
 (iii) M = Methods of production
 (iv) p_i = Price of machines and raw material.

Thus $qs = f(p, c, m, p_i)$

To make our study simple, it is assumed that except price of the good (p) all other factors remain constant in the short period.

Thus supply directly depends upon price which may be written as under:

$qs = f(p)$. Including a and b parameters in the supply function, functional equation may be shown as under:

$$qs = ap + b \quad \text{or} \quad qs = b + ap$$

Putting the values of $a = 1$ and $b = 4$

$$q = p + 4$$

GRAPHIC REPRESENTATION OF SUPPLY EQUATION

If $qs = 4 + p$

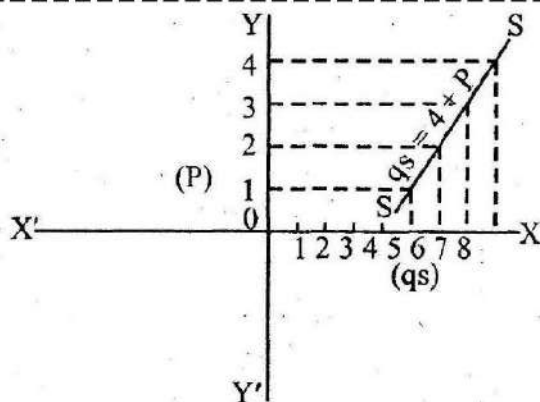
Here p is an independent variable. By giving assumed values to $p=1, 2, 3, 4$ we can calculate the values of qs in the above equation.

(i)	$qs = 4 + p = 4 + 1 = 5$
(ii)	$qs = 4 + p = 4 + 2 = 6$
(iii)	$qs = 4 + p = 4 + 3 = 7$
(iv)	$qs = 4 + p = 4 + 4 = 8$

Putting these values in the schedule

P	1	2	3	4
qs	5	6	7	8

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Description

Along ox quantity supplied and with oy price is shown. Keeping in view various combinations of p and qs. SS supply curve has been drawn. It moves upwardly showing positive relationship between p and qs. It represents the equation $q = 4 + p$

DERIVATION OF SUPPLY EQUATION

We can get a functional equation of supply from the given supply schedule with the help of following formula.

$$q - q_1 = \frac{q_2 - q_1}{p_2 - p_1} (p - p_1)$$

Here

q = symbol of quantity supplied

q_1 = original quantity

q_2 = New quantity

p = symbol of price

p_1 = original price

p_2 = New price

Example: Find functional equation of supply from the following scheme.

	p_1	p_2	
p	1	2	3
q	10	15	20
	$\uparrow q_1$	$\uparrow q_2$	

By putting values in the formula:

$$q - 10 = \frac{15 - 10}{2 - 1} (p - 1) = \frac{5}{1} (p - 1) = 5p - 5$$

$$q - 10 = 5p - 5, q = 5p - 5 + 10$$

$$q = 5 + 5p$$

$$q = 5 + 5(1) = 10 \text{ while } p = 1$$

$$q = 5 + 5(2) = 15 \text{ while } p = 2$$

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CHANGES IN SUPPLY

Changes in supply may be discussed as under:-

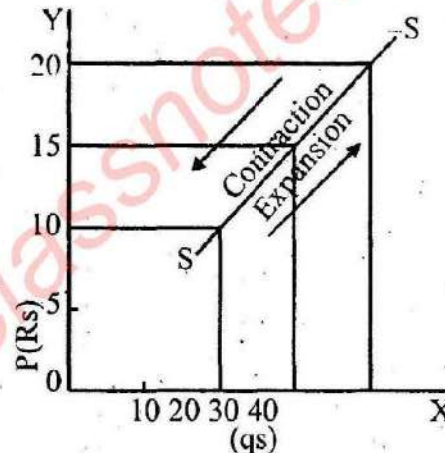
1. MOVEMENT ALONG THE SUPPLY CURVE OR EXPANSION AND CONTRACTION IN SUPPLY

When the quantity supplied increases in view of an increase in price of the commodity it is known as an "expansion" or "extension" in supply. On the other hand, when supply decreases due to decrease in the price of good. It is regarded as "contraction" in supply. Here supply changes only due to changes in price of the commodity concerned.

Under these conditions, the supply schedule and curve remain the same. The upward movement along the supply curve shows "expansion" in supply while, the downward movement indicates "contraction" of supply. It is clear from the following schedule and diagram.

p(Rs)	qs(K.G)
10	20
15	30
20	40

Contraction ↑ Expansion ↓



An upward trend of arrow in diagram shows "Expansion in Supply" while the downward trend indicates the "contraction" of supply. This is also known as the movement along the same supply schedule and curve.

2. SHIFT IN SUPPLY CURVE OR RISE AND FALL IN SUPPLY.

(a) Rise in Supply:

When supply of a commodity increases not due to change in its own price but due to change in "other factors" which are assumed to remain constant e.g. C.M. P_i etc. It is known as "Rise in supply." Under these conditions New Supply Curve shifts downward to the right of original supply curve.

This downward shift of supply curve may be shown under two conditions.

- Rise in quantity supplied at the same price.
- Quantity supplied remains the same at lower price.

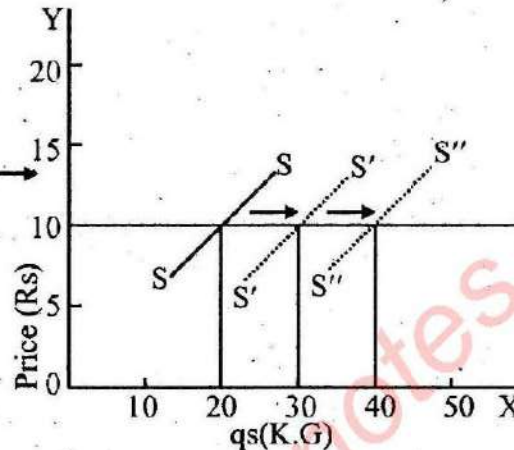
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RISE IN SUPPLY

SCHEDULE

(i) More qs at same price

p(Rs)	qs(K.G)
10	20
10	30
10	40

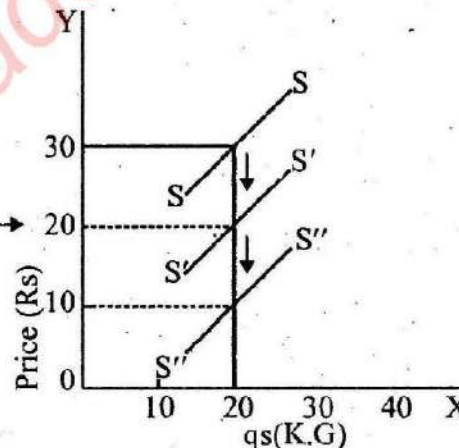


Explanation

SS is the original supply curve showing combination of P. (Rs.10) and qs (20) in the schedule. While price remaining the same with more supply (30) S'S' moves downward right to the original supply curve SS. While further rise in qs (40) at the same price helps to shift SS downward to S''S''.

(ii) Same qs at lower price

p(Rs)	qs(K.G)
30	20
20	20
10	20



ExplanationP

SS is the original supply curve. When price goes down (Rs.20) without any change in qs, S'S' new supply curve moves downward. Similarly, with the further decrease in price (Rs.10) S''S'' shifts downwardly showing a "rise in supply." Above mentioned "two ways" indicate "rise in supply".

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FALL IN SUPPLY

When quantity supplied of a commodity decreases at the same price or rise in price does not affect supply (due to the pressure of other factors assumed to remain constant) is known as "fall in supply".

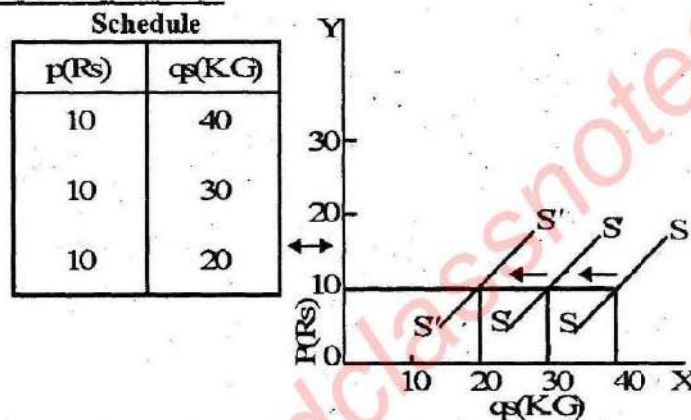
Under these conditions new supply curve shifts "upward as $S'S'$ and further as $S''S''$ ".

It can be explained as under:-

- (i) Decrease in quantity supplied at the same price.
- (ii) Quantity supplied remains the same at higher price.

FALL IN SUPPLY

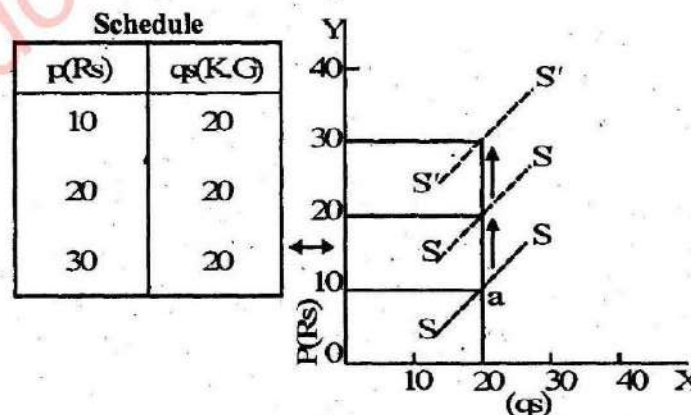
(I) LESS QS AT SAME PRICE



EXPLANATION

SS is the original supply curve showing combination of P (Rs.10) and qs (40) at point (a). Price remaining the same with less qs (30) $S'S'$ moves upward to left of the original SS. While further decrease in qs (20) at the same price results a shift of $S''S''$ more upward to the left of $S'S'$ at point (C).

(ii) Same qs at higher price.



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Explanation:

SS is the original supply curve showing combination of P(Rs.10) and q_s (20) at point (a). Q_s remains the same even though price rises to (Rs.20). New supply curve $S'S'$ shifts upward. More increase in P(Rs.30) at same q_s (20) results a further upward shift in supply curve as $S''S''$.

These two ways indicate "fall in supply".

CAUSES OF CHANGES IN SUPPLY

Following factors are mainly responsible for changes in supply.

1. CHANGE IN THE PRICE OF GOOD:

As $q_s = f(p)$ thus any change in the price of good brings change in its supply in the same direction.

2. CHANGE IN THE COST OF PRODUCTION:

Supply falls with the increase in the cost of production of the good while, on the other side, a decrease in the cost of production, may cause rise in its supply.

3. CHANGE IN PRODUCTION TECHNIQUES

Supply may rise due to use of latest technology and modern techniques of production e.g. use of tractors, harvesters etc in agriculture. Use of automatic machines and computers in production process is also helpful for more production and increase in supply.

4. CHANGE IN THE PRICE OF MACHINES AND RAW MATERIAL

Supply falls when prices of machines, raw material and other inputs are increased. It causes a rise in the cost of production leaving a very small profit margin, on the other hand, supply may rise when these factors are available at cheaper rates.

5. CHANGE IN POLITICAL CONDITIONS.

Business expands and supply rises under stable and peaceful political conditions. While political in-stability and un-rest in the country discourages investment and production activities. Thus a fall can be seen in the quantity supplied of goods in the market.

6. NATURAL CALAMITIES

Supply of goods particularly in Agriculture sector is badly affected due to natural calamities e.g. Floods, Soil erosion, Scarcity of rain, Crop diseases etc. War can also reduce supply.

7. CHANGE IN TAXES

An increase in Tax rate or impact of new taxes on commodities may cause fall in supply. Sales tax, excise duty and agricultural tax shift supply curve upward while Tax reduction, Tax concession and Tax holidays encourage production activities resulting a rise in supply.

8. DEVELOPMENT IN THE MEANS OF TRANSPORTATION

Supply may rise due to improved transport and communication facilities. Because the shifting of goods become cheaper and easier, market expands, demand increases and supply rises.

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9. WEATHER CONDITIONS.

Supply also rises or falls according to the changing weather conditions. Supply of ice falls during winter season while it rises during summer.

10. PRICES OF SUBSTITUTES

Change in price of substitute can affect the supply of related good. For example, a rise in the price of wheat may encourage the supply of rice.

11. UNITY AMONG PRODUCERS

This type of unity may be in the form of cartels. Trust, price leadership, supply quota etc. Under such a situation supply may fall. Sometime producers intentionally reduce production and supply of certain goods so that prices of these goods may not go down beyond a certain limit.

12. FUTURE EXPECTATIONS

If entrepreneurs foresee some changes in future prices. They try to adjust their output accordingly and take the decisions to increase or decrease the supply of goods in view of the future expected prices.

SOME OTHER CONCEPTS OF SUPPLY

1. COMPOSITE SUPPLY

Sometime there are different sources for the supply of a good. When we take up the sum total of supply from all the available resources, we arrive at "Composite Supply" of the good e.g. meat from sheep, goat, lamb etc. and beef from cow, bull and buffalo. Similarly pots are prepared from steel, Bronze, plastic, mud etc. As these goods have "Composite Supply" Thus their prices are interlinked with each other.

2. JOINT SUPPLY

Some goods are jointly produced. Thus an increase in the supply of one commodity definitely results a rise in the supply of the other good and vice versa. These are known as "Jointly supplied goods". E.g. Wheat with straw, meat with wool and leather from a sheep etc.

3. COMPETITIVE SUPPLY

Supply of some goods compete with each other when there are more than one uses. If the supply is increased in one use, its supply for the other use will become less. For instance, if supply of electricity is increased for house hold purposes its supply may be decreased for commercial and industrial uses.

ELASTICITY OF SUPPLY

Elasticity of supply is also a very important concept in microeconomic theory. This term is known as "the degree of responsiveness of quantity supplied to the change in the price of a commodity."

When a minor change in price causes a large change in supply of goods. These are known as "more elastic supplied goods" on the other hand. When supply shows small. Changes in response to a big change in price. The supply is regarded as "less elastic".

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MEASUREMENT OF ELASTICITY OF SUPPLY

Major methods in this respect are as under:-

1. UNITY METHOD

According to this method elasticity of supply may be:-

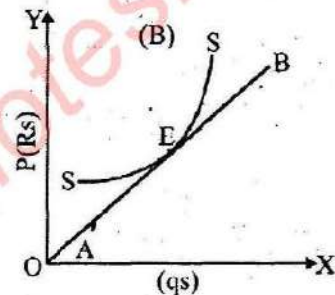
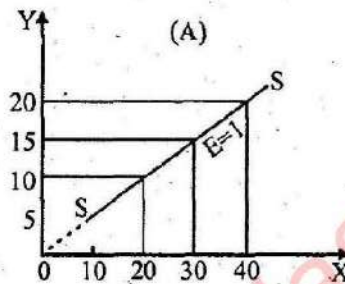
- (a) $E = 1$ (Equal to unity)
- (b) $E > 1$ (Greater than unity)
- (c) $E < 1$ (Less than unity)

a. $E_s = 1$ (Equal to unity)

When proportionate change in quantity supplied and price is equal or the rate of change in both remains the same, $E_s = 1$

Schedule

p(Rs)	qs(K.G)
10	20
15	30
20	40



Explanation

In diagram (A), SS (supply curve) is a straight line with positive slope. It passes through origin (0) showing an equal distance from x-axis and y-axis, thus $E_s = 1$.

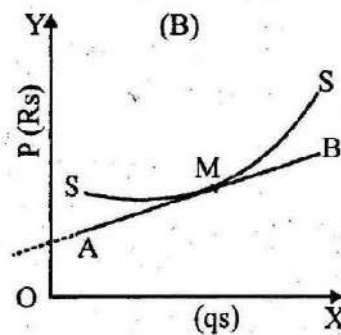
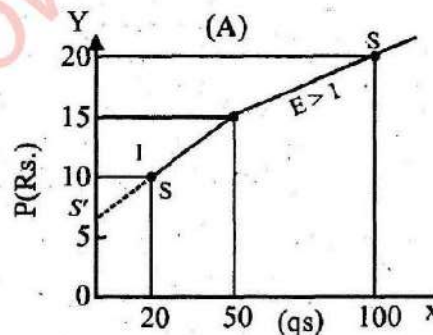
In diagram (B) SS is in the shape of a curvature. Its tangent AB at point E, also passes through (0) showing $E_s = 1$, at this point.

(b) $E_s > 1$ (Greater than unity)

If rate of change in supply is greater than the rate of change in price of a commodity. It shows that $E_s > 1$.

Schedule

p(Rs)	qs(K.G)
10	20
15	50
20	100



Explanation:

In diagram (A) SS is a flatter straight line supply curve. When extended towards left it crosses y-axis showing that $E_s > 1$.

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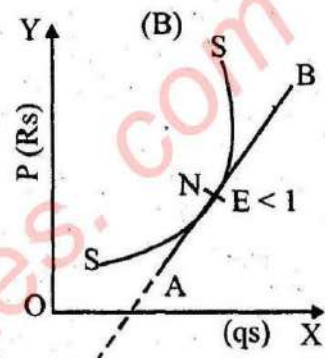
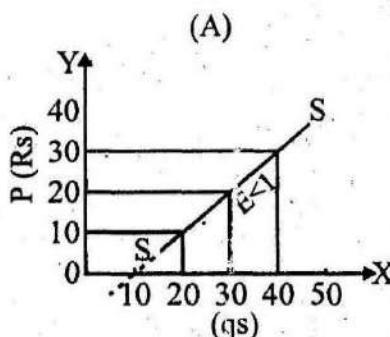
In diagram (B) SS is in the form of a curvature. It's tangent AB at point M also exhibits that being flatter at this point $E_s > 1$.

(c) $E < 1$ (less than unity)

When proportionate change in quantity supplied is smaller than proportionate change in price. It shows that $E_s < 1$, under such a situation SS curve becomes more steeper.

Schedule

p(Rs)	qs(K.G)
10	20
20	30
30	40



Explanation:

In diagram (A) Ss is a steeper straight line supply curve, when extended towards left it crosses x-axis showing that any combination at this curve indicates $E_s < 1$.

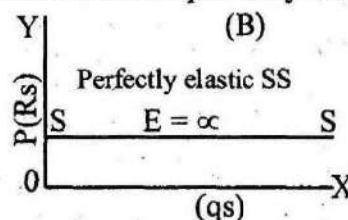
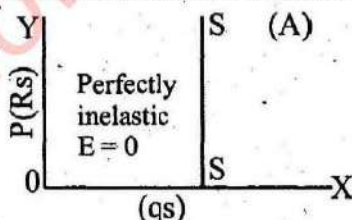
In diagram (B) Ss is in curvature form, AB is its tangent touching at point N. As AB crosses x-axis when extended towards left thus at point N, $E_s < 1$.

Hence according to the unity method:

$$E_s = \frac{\text{Proportionate change in quantity supplied}}{\text{Proportionate change in supply}}$$

Note: Two extreme cases:

- Elasticity of supply may become perfectly inelastic ($E_s = 0$) when supply does not respond to any change in price. SS curve becomes vertical upon x-axis.
- $E_s = \infty$ (infinity) when supply goes on increasing at a given price of the commodity. SS curve becomes horizontal towards x-axis which is perfectly elastic.



2. PERCENTAGE METHOD:

Under this method % change in quantity supplied and price is kept before as:-

$$E_s = \frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}$$

- When there is an equal % change in price and quantity supplied then $E_s = 1$.
- When % change in quantity supplied is greater than the % change in price $E_s > 1$.

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(iii) When % change in supply is less than the % change in price $E_s < 1$.

3. FORMULA METHOD

Supply elasticity of any commodity may be measured with the help of following formula:

$$E_s = \frac{\frac{\Delta q}{q}}{\frac{\Delta p}{p}} = \frac{\Delta q}{q} \div \frac{\Delta p}{p} = \frac{\Delta p}{p} \times \frac{p}{\Delta p} = \frac{p}{q} \cdot \frac{\Delta q}{\Delta p}$$

E_s = Symbol of supply elasticity

q = Original quantity supplied, Δq = change in supply

p = Original price, Δp = change in price.

Example:

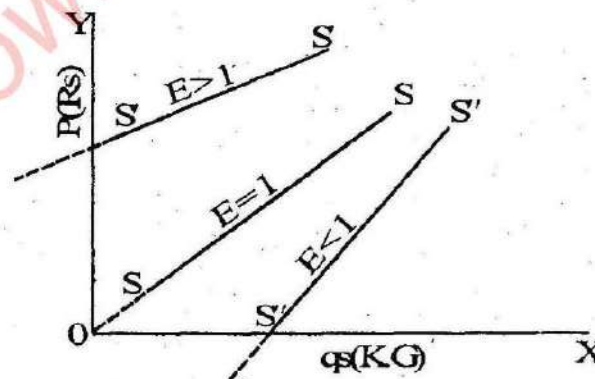
	P(Rs)	qs(K.G)	
P →	10	20 ← q	
	12	24	
ΔP →	2	4 ← Δq	

Putting the values in the above formula:

$$E_s = \frac{\frac{4}{20}}{\frac{2}{10}} = \frac{4}{20} \times \frac{10}{2} = \frac{1}{5} \times \frac{5}{1} = 1$$

Thus here $E_s = 1$

Whole discussion with respect to the Elasticity of supply may be summarized as under:



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FACTORS INFLUENCING THE ELASTICITY OF SUPPLY

Following factors are known as the determinants of supply elasticity.

1. NATURE OF GOODS:

Perishable goods are inelastic because these cannot be stored. On the other hand as durable goods may be stored for a comparatively longer period, thus these have more elastic supply.

2. CHANGES IN PRODUCTION TECHNIQUES

In the presence of advanced technology new discoveries and modern methods of production, goods may be produced at cheaper rates. Supply can easily be increased thus these have "more elastic supply".

3. APPLICATION OF THE LAWS OF RETURN:

Production process of goods passes through different stages:

Under "law of increasing return" increase in production is profitable. Hence supply is "more elastic".

On the other side, supply of a good becomes "less elastic" under the "law of decreasing return". As marginal product goes on decreasing and additional product becomes a little profit oriented.

4. PRODUCTIVE CAPACITY OF INDUSTRY

If an industry is working in "full capacity". It becomes difficult to increase production particularly, in short-period. Thus supply is "less elastic".

On the other hand, when an industry has "reserve capacity" and is working below full capacity, supply can easily be increased and thus supply becomes "more elastic".

5. TIME PERIOD

In day to day market "supply is in-elastic". In the short run supply becomes "less elastic". It is "more elastic" in the long period because time is sufficient to make adjustment of supply according to the demand.

6. CAPITAL FORMATION

If the speed of capital formation is according to the investment requirements, the production of goods may be increasing accordingly and supply will be "more elastic". On the other hand a low rate of capital formation will result "less elastic supply of goods".

7. OTHER FACTORS

A restriction upon imports or exports, backward means of transportation and decrease in production. May result "less elastic supply" On the other side free trade policy, developed means of transportation and large extent of market causes "more elastic supply".

PRACTICAL USES OF THE CONCEPT

1. STABILITY OF PRICES:

Elasticity helps to adjust supply according to the changes in demand. Equilibrium between the two keeps stability in prices.

2. AVAILABILITY OF GOODS ACCORDING TO REQUIREMENTS

As many goods are elastic in supply. Thus these are available according to the requirements. The question of scarcity can not arise.

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3. KNOWLEDGE ABOUT THE "PRODUCTIVE CAPACITY" OF INDUSTRY

More or less elastic supply indicates whether industry is working in "full capacity" or "under reserve capacity". To increase production of the goods "reserve capacity" may be fully utilized.

4. HELPFUL FOR STOCK OF GOODS

durable goods may be kept in the stock because these goods are more elastic in supply" on the other hand, perishable goods being inelastic in supply cannot be stored for a longer period.

5. A GUIDELINE FOR TAXATION

Supply elasticity of goods helps to impose taxes. More elastic goods cannot be taxed heavily. Because it affects demand for goods adversely. While more revenue can be collected through heavy taxes levied upon goods being in-elastic or less elastic in supply.

OBJECTIVE QUESTIONS

Select and tick (✓) Correct answer.

1. When supply rises:
(a) Supply curve does not move any way. (b) It become horizontal to y-axis.
(c) It moves downward to right. (d) It moves upward to left.
2. Under In-elastic supply:
(a) Supply curve becomes horizontal to x-axis.
(b) Supply curve becomes horizontal to y-axis
(c) Supply curve rises upwardly
(d) Supply curve falls downwardly.
3. Total quantity of wheat (500 bags) lying in a store to be supplied in future at the required rate is known as:
(a) Supply (b) Production (c) Stock (d) Buffer stock
4. Keeping in view the following information the elasticity of supply will be
(a) $E = \infty$ (b) $E > 1$
(c) $E = 1$ (d) $E < 1$

P(Rs)	Supply (K,G)
20	100
22	110
5. Find out the supply equation as under:
(a) $q + 3p - 6 = 0$ (b) $q = 5 - 0.5 p$
(c) $3p - q = 12$ (d) $q - 1200 - 4p$
6. Supply of a good depends upon..?
(a) Utility (b) Will (c) Price (d) Fashion
7. What type of relationship supply has with price of good.
(a) Inverse (b) Negative (c) No relationship (d) Positive
8. What causes shift in supply curve.
(a) Other factors (b) Change in price
(c) Change in demand (d) Fiscal policy
9. If Govt. imposes restrictions upon imports and exports of goods, the supply will become...?
(a) More elastic (b) Less elastic (c) In-elastic (d) Infinite elastic

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10. Slope of perishable goods supply curve becomes...?
(a) Horizontal (b) Vertical (c) Negative (d) Exceptional
11. Which will be regarded as quantity supplied...?
(a) Sugar in the market (b) Wheat in ware house
(c) Onion in the field (d) Potatoes in cold storage.
12. Why supply curve is positively sloped...?
(a) Demand for cash money (b) Increase in profit
(c) Threat of war (d) Obstacles in the way of transportation.
13. Supply of wheat and straw is..?
(a) Composite (b) Joint (c) In-elastic (d) Competitive
14. When supply remains unchanged even if price rises, then it is regarded....?
(a) Fall in supply (b) Rise in supply
(c) Extension in supply (d) Contraction in supply
15. Supply curve of "fish" will be....?
(a) More elastic (b) Infinite elastic (c) In-elastic (d) Flatter
16. What is a "News paper".
(a) Durable good (b) Most costly good
(c) Rare good (d) Perishable good
17. If price increases by 10% and supply expands by 15% the elasticity of supply will be:
(a) $E < 1$ (b) $E = 1$ (c) $E > 1$ (d) $E = 0$
18. In view of time period supply is of?... kinds...?
(a) One (b) Two (c) Three (d) Four
19. When the supply of electricity falls for commercial and industrial units due to rise for household consumption, it will be regarded as..?
(a) Composite supply (b) Competitive supply
(c) Joint supply (d) Essential supply
20. Which one of the following is not included among determinants of supply.
(a) Consumer's choice (b) Cost of production
(c) Methods of production (d) Price of raw material.

DESCRIPTIVE PART - I

SHORT NOTES

1. What is meant by Supply?

Supply is the quantity of a commodity offered for sale in the market at a specific price level during, a specific period.

2. What is meant by "Stock"?

Stock represents the "total quantity produced" of a commodity available in shops, houses farm-houses or warehouses near the market which can be brought for sale at short notice.

3. Define the "Law of Supply"?

Other things remaining the same, quantity supplied of a commodity expands with an increase in its price and contracts with the decrease in price. $Q_s = f(P)$ and it is an increasing function.

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4. Write down four assumptions of law of supply?

- 1) No change in cost of production. (C)
- 2) No change in production techniques. (M)
- 3) No change in price of machines and raw material. (P_1)
- 4) No government restrictions. (GR)

5. Point out the limitations of law of supply?

Intensive need of cash, money, un-certain conditions, Immediate immigration, lac of transportation facilities etc.

6. Discuss various types of supply in view of time period?

Fixed supply, short period supply, Long period supply

7. What is composite supply?

The sum of total supply from different sources is called composite supply e.g. meat from sheep, goat, lamb or energy from gas, oil, electricity, hydel power etc.

8. Explain the concept of joint supply?

Some goods are jointly produced e.g. wheat with straw, meat with wool, leather from sheep if the supply of one commodity increases, results in an increase in other goods, supply, and vice versa.

9. Point out three factors of elasticity of supply?

- 1) Nature of good.
- 2) Changes in production techniques
- 3) Application of law of return

10. How Elasticity of Supply (ES) is measured according to unitary method?

According to unitary method elasticity of supply may be of three types.

- a) $E=1$ equal to unity
- b) $E<1$ less than unity
- c) $E>1$ greater than unity

11. What is functional equation of supply?

Keeping in view the determinants the equation is $q_s = f(p, c, m, p_i)$ assuming other factors constant except price, equation will be $q_s = f(P)$. other factors are C, M, P_i .

12. Explain the concept of zero elasticity of supply?

When supply does not respond to any change in price and becomes inelastic, in such case supply elasticity becomes zero. The supply curve becomes vertical towards X-axis.

PART – II

1. Explain the "law of supply" with the help of a schedule and diagram.
2. Distinguish between expansion or contraction and rise or fall in supply.
3. How elasticity of supply is measured.
4. Under which circumstances shifting of supply curve is possible. Explain with the help of diagram.
5. Give graphic representation of the following supply equations.
(i) $q - p = 20$, (ii) $q = -2 + 6p$
6. Derive supply equation with the help of following data.

p	5	3.5
q	5000	2000

